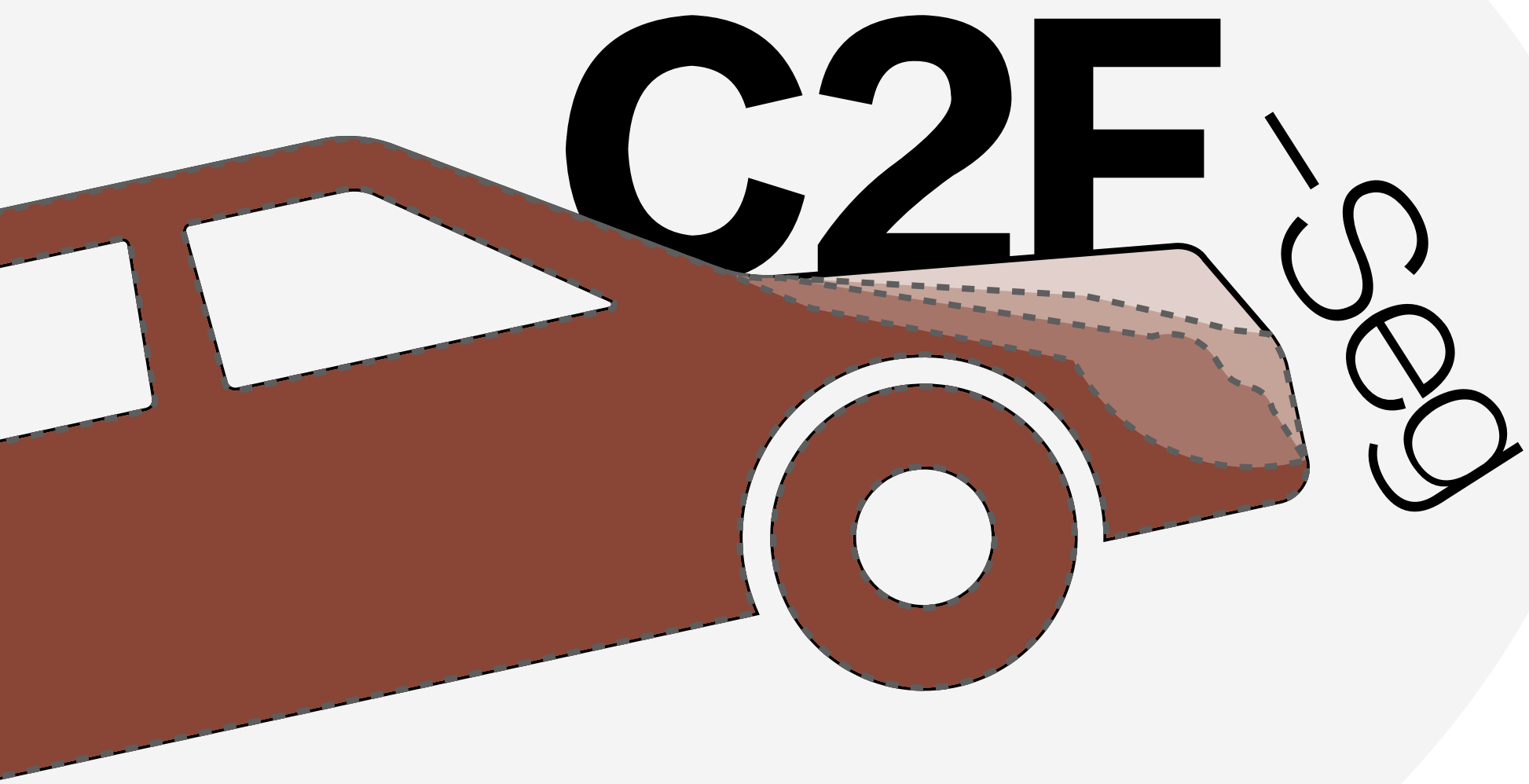




Coarse-to-Fine Amodal Segmentation with Shape Prior

ICCV2023





1 向量化

1 向量化



Visible Mask $\widehat{M}_v$



Input Image I

1 向量化



Visible Mask $\widehat{M}_v$



$v_{\widehat{M}_v}$



Input Image I



v_{img}

1 向量化



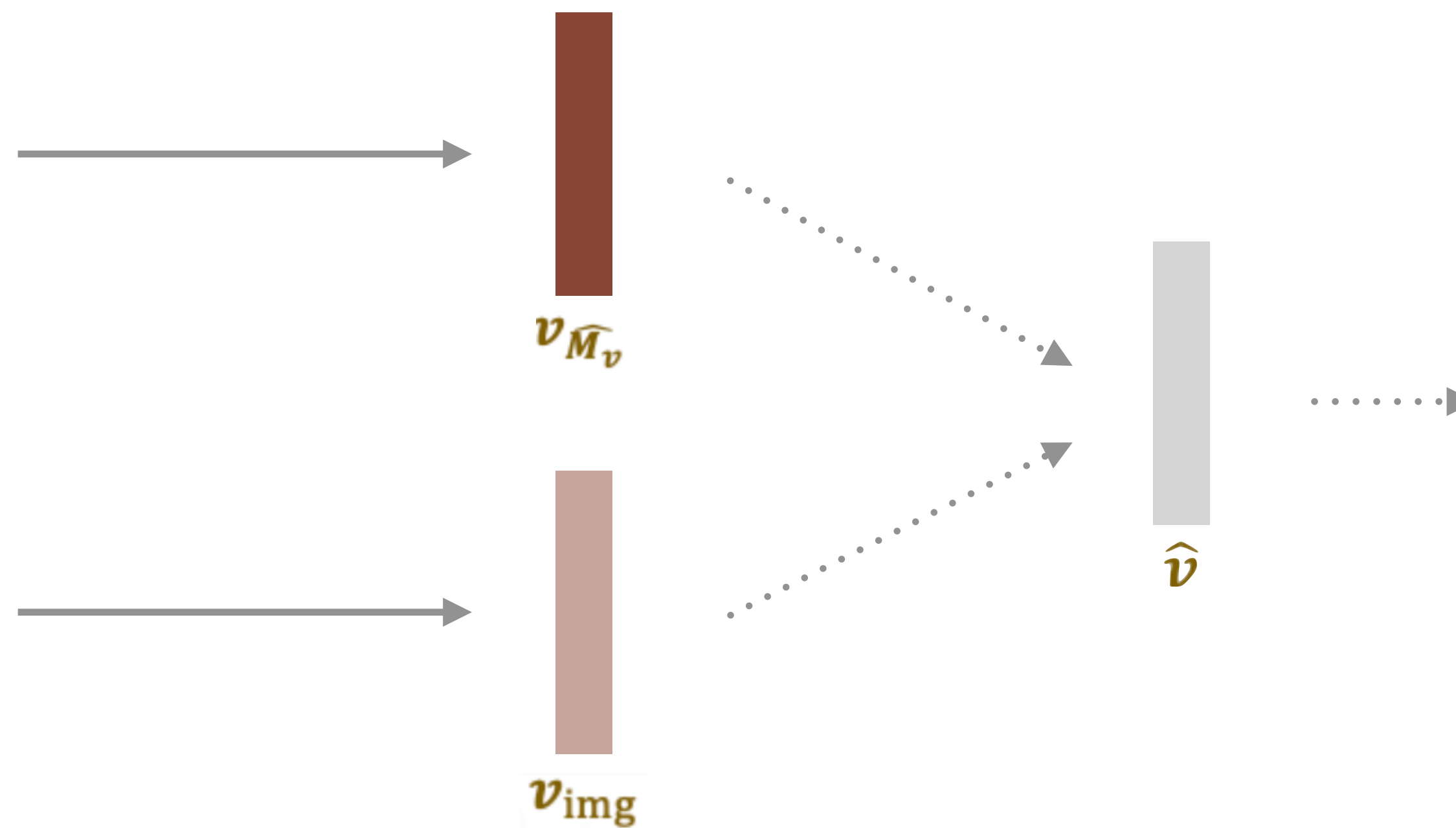
Visible Mask $\widehat{M}_v$



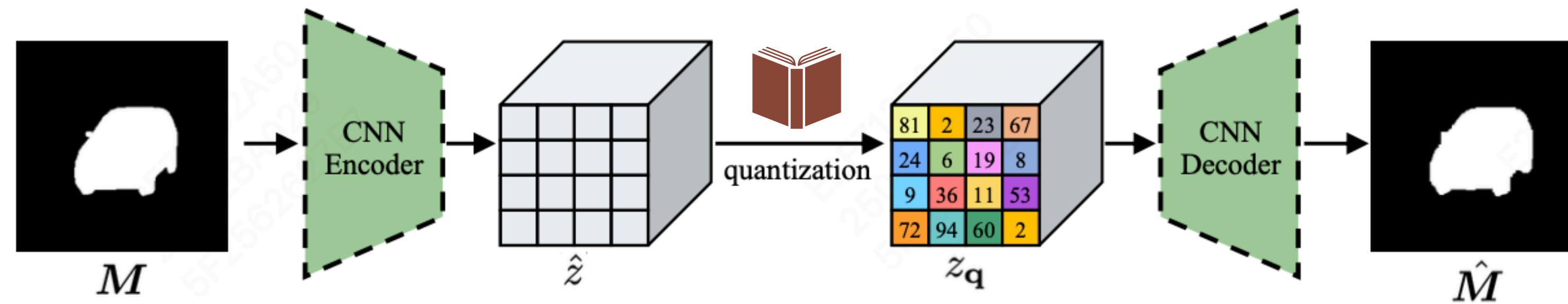
Input Image I



Input Image I



1 向量化

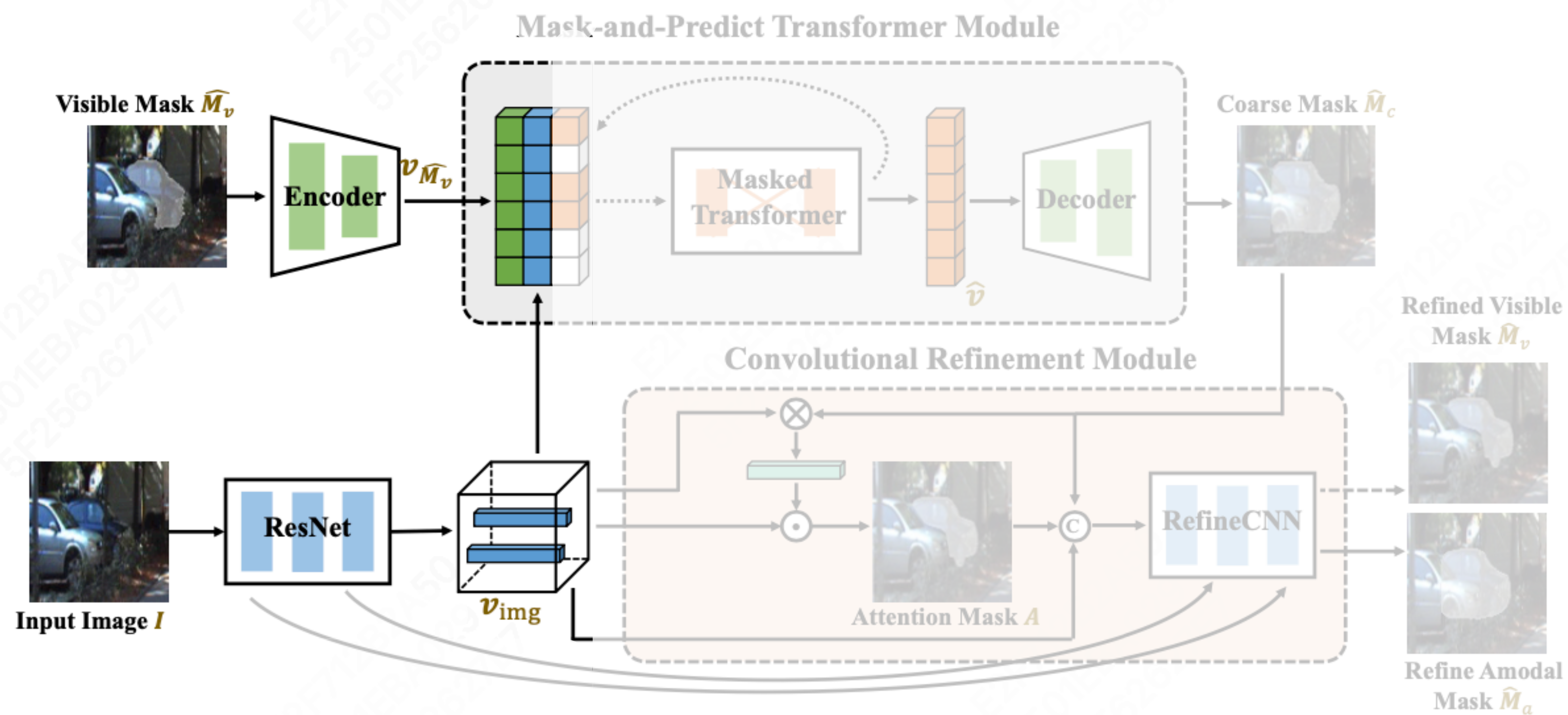


Vector-Quantization Model

$$z_q = \mathbf{q}(\hat{z}) := \left(\arg \min_{z_k \in \mathcal{Z}} \|\hat{z}_{ij} - z_k\| \right) \in \mathbb{R}^{h \times w \times n_z}$$

$$v_M := \text{Embed}(s). \quad s \in \{0, \dots, |\mathcal{Z}| - 1\}^{h \times w}$$

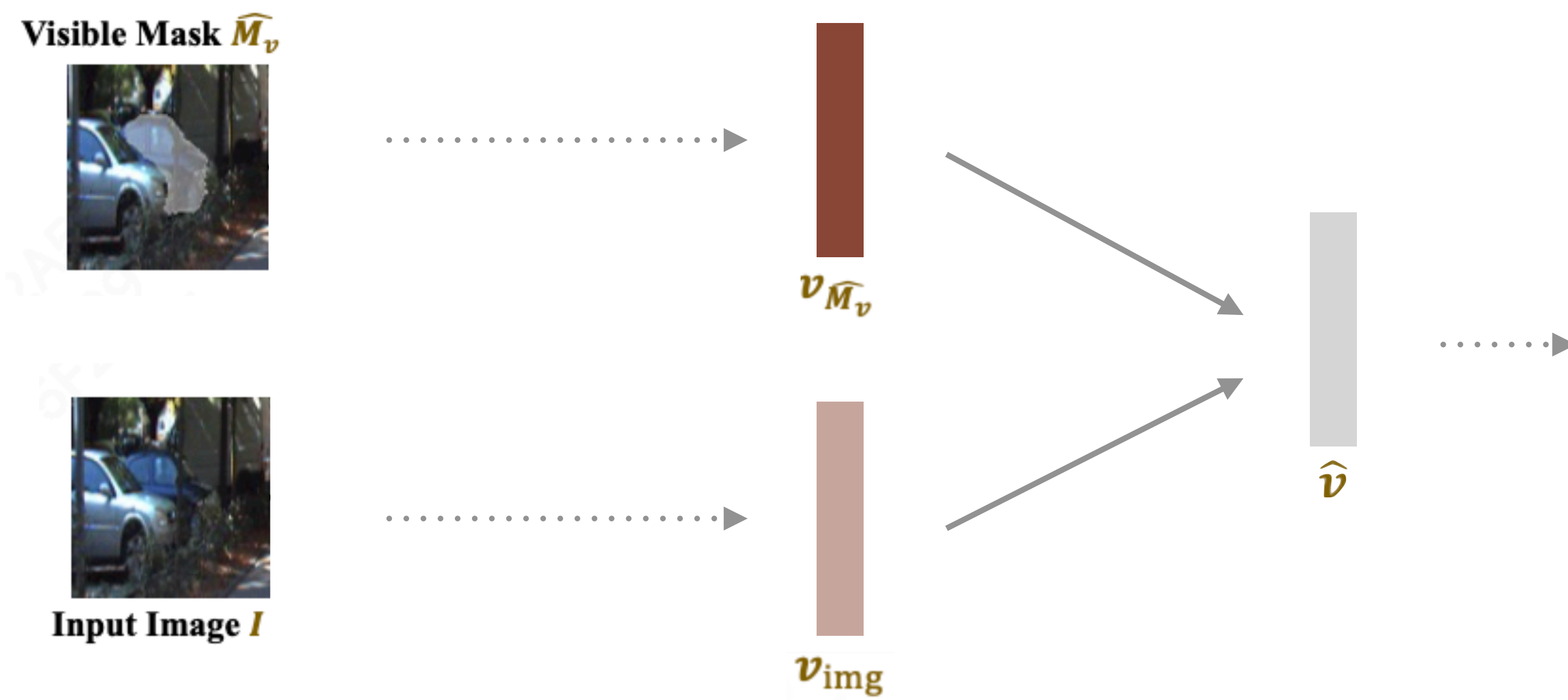
1 向量化



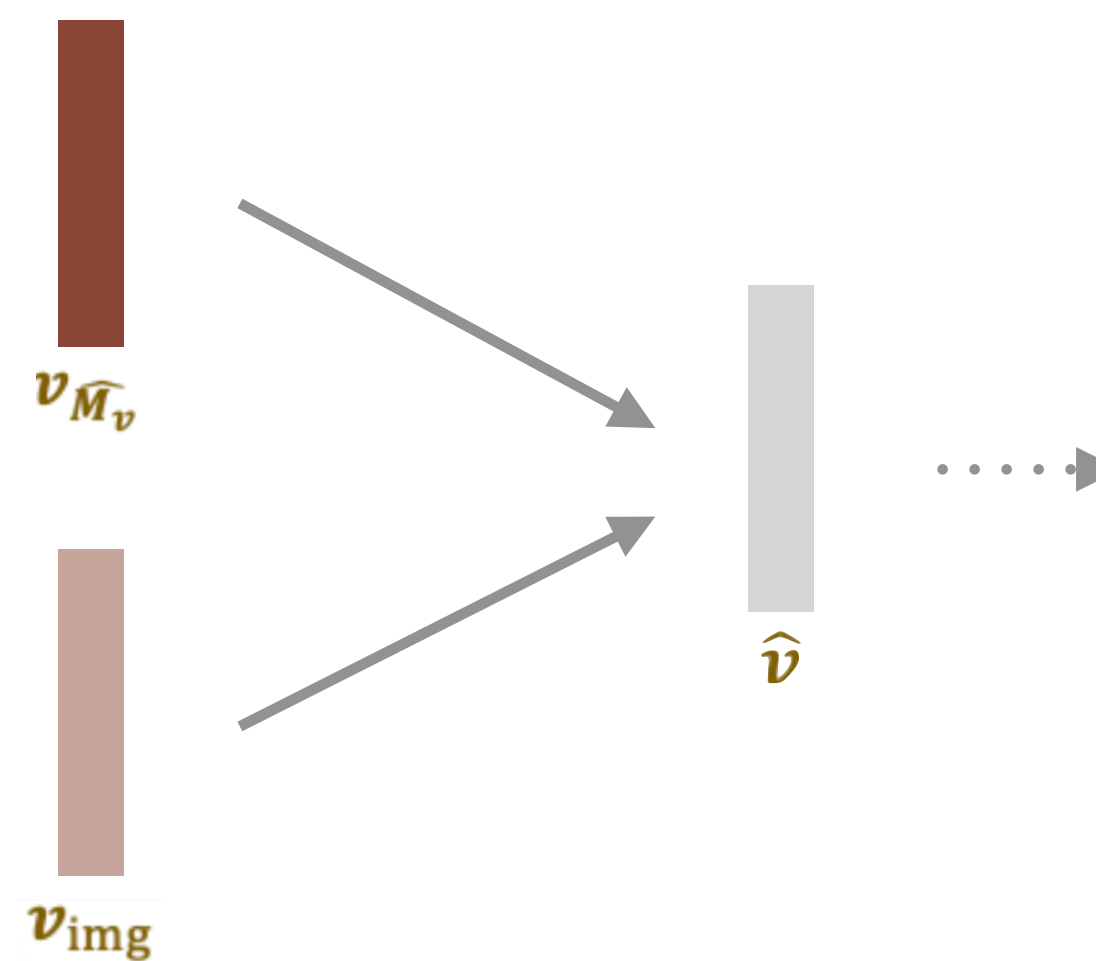


2 转换

2 转换



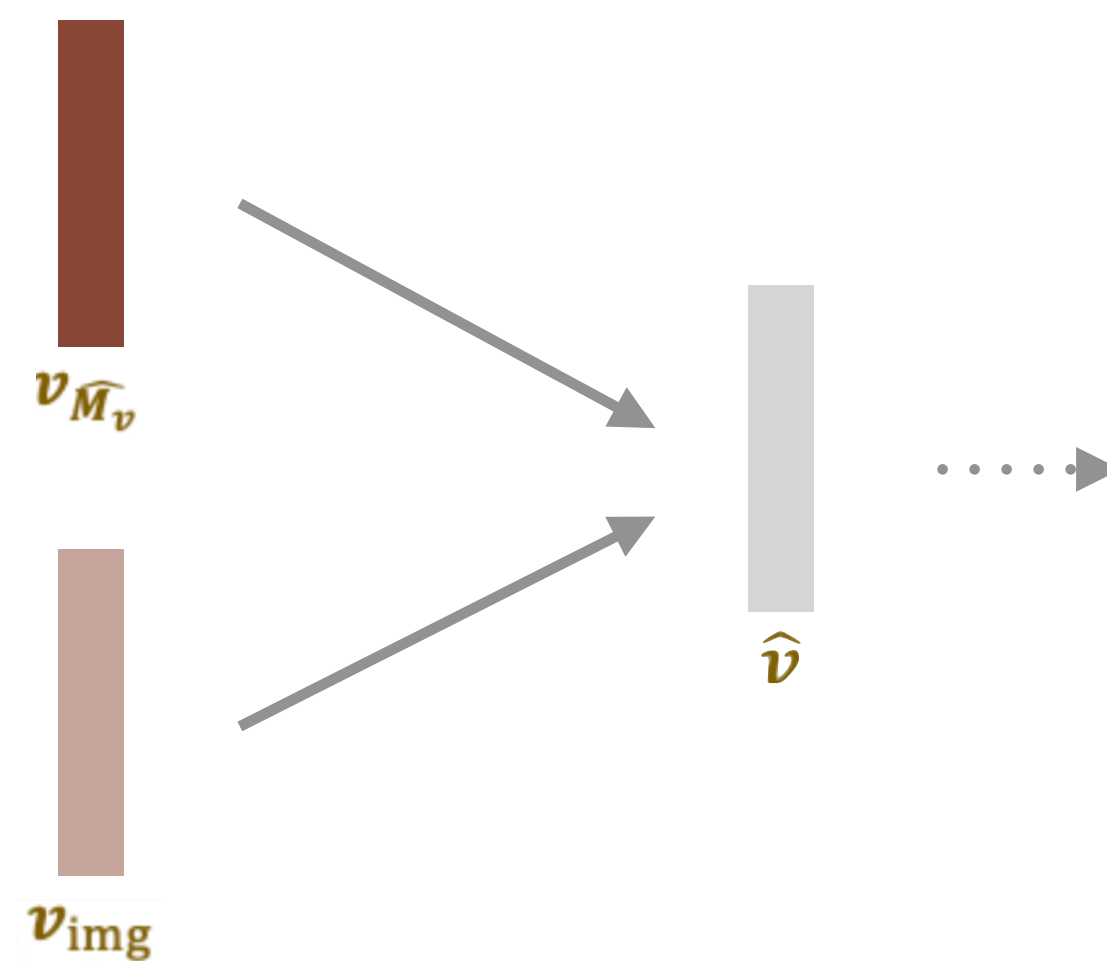
2 转换



极大似然法

$$\mathcal{L} := -\mathbb{E} \left[\sum_i \log p(\mathbf{v}_{M_a, i} \mid \hat{\mathbf{v}}, \mathbf{v}_{\text{img}}, \mathbf{v}_{\hat{M}_v}) \right]$$

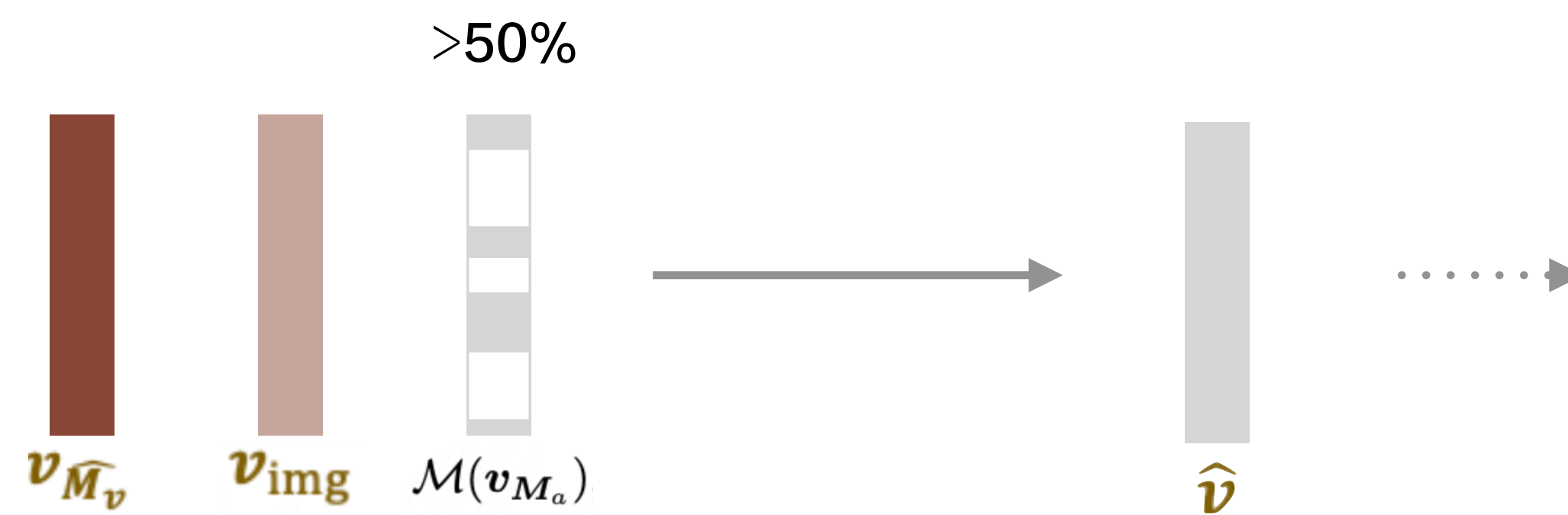
2 转换



极大似然法

$$\mathcal{L} := \mathbb{E} \left[\sum_i \log p(\mathbf{v}_{M_a, i} | \hat{\mathbf{v}}, \mathbf{v}_{\text{img}}, \mathbf{v}_{\hat{M}_v}) \right]$$
$$\mathcal{L} := -\mathbb{E} \left[\sum_i \log p(\mathbf{v}_{M_a, i} | \mathcal{M}(\mathbf{v}_{M_a}), \mathbf{v}_{\text{img}}, \mathbf{v}_{\hat{M}_v}) \right]$$

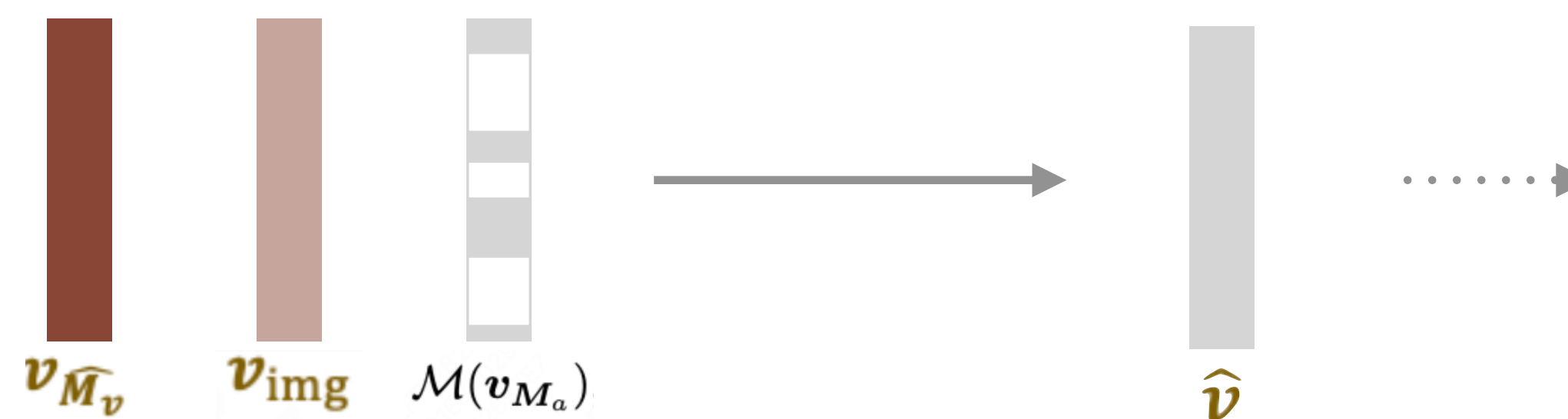
2 转换



训练

$$\mathcal{L} := -\mathbb{E} \left[\sum_i \log p(v_{M_a,i} \mid \mathcal{M}(v_{M_a}), v_{\text{img}}, v_{\hat{M}_v}) \right]$$

2 转换



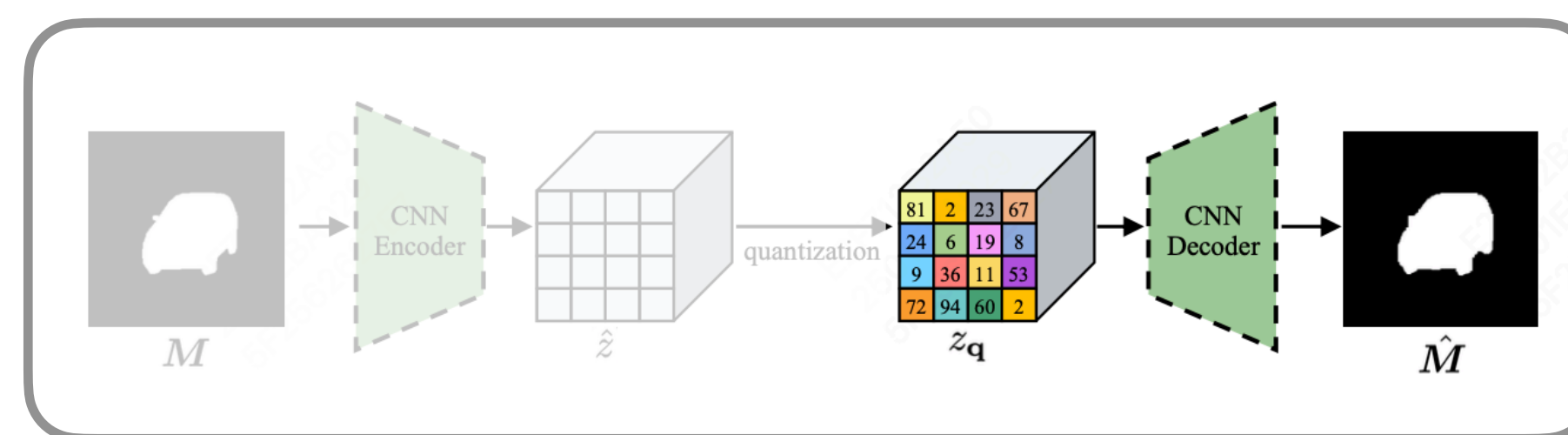
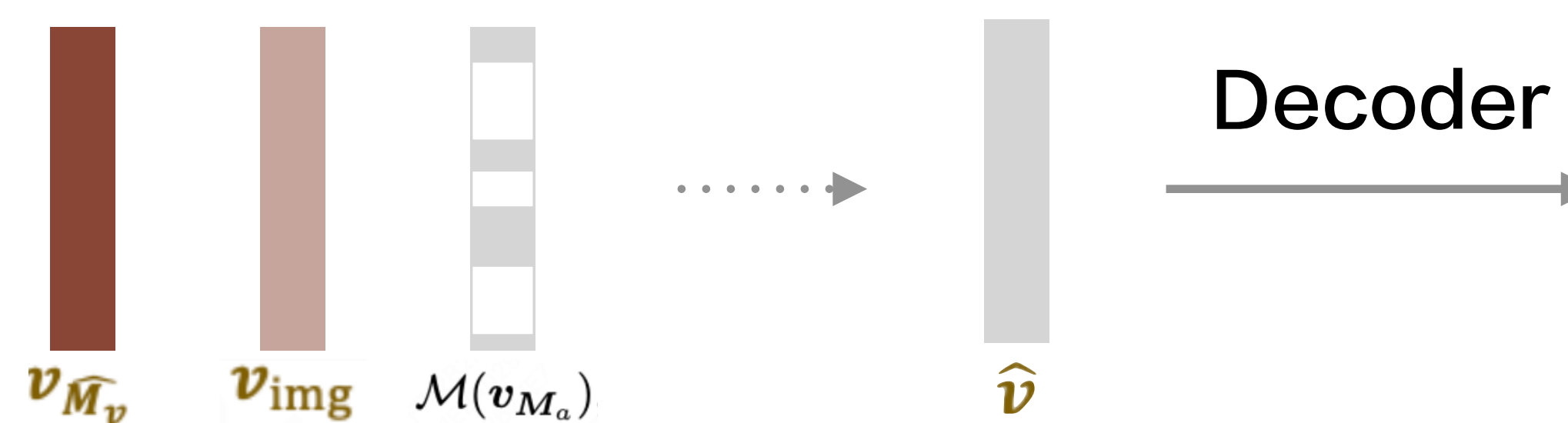
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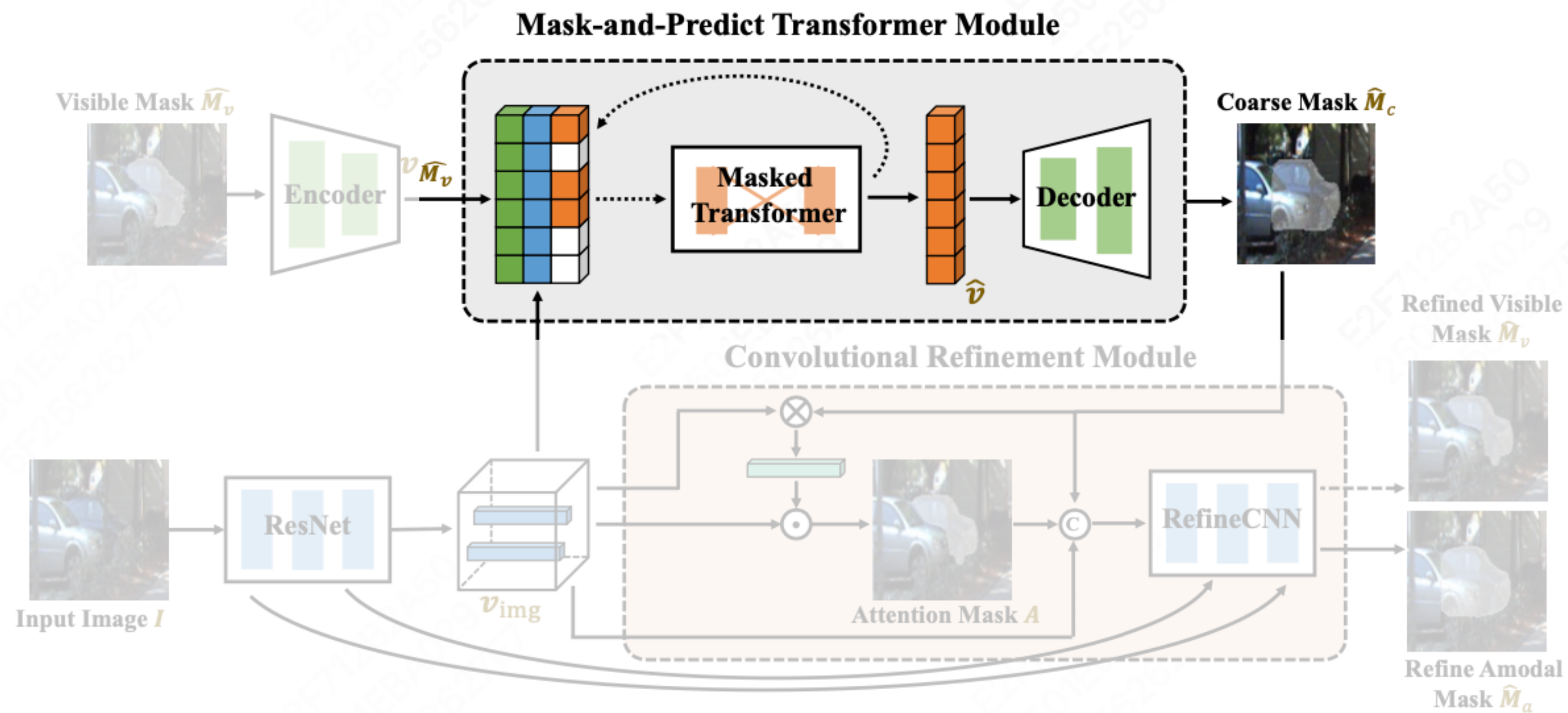
推理

迭代推理，每次采纳置信度达到阈值的项，直至完全或到达 K 次

2 转换



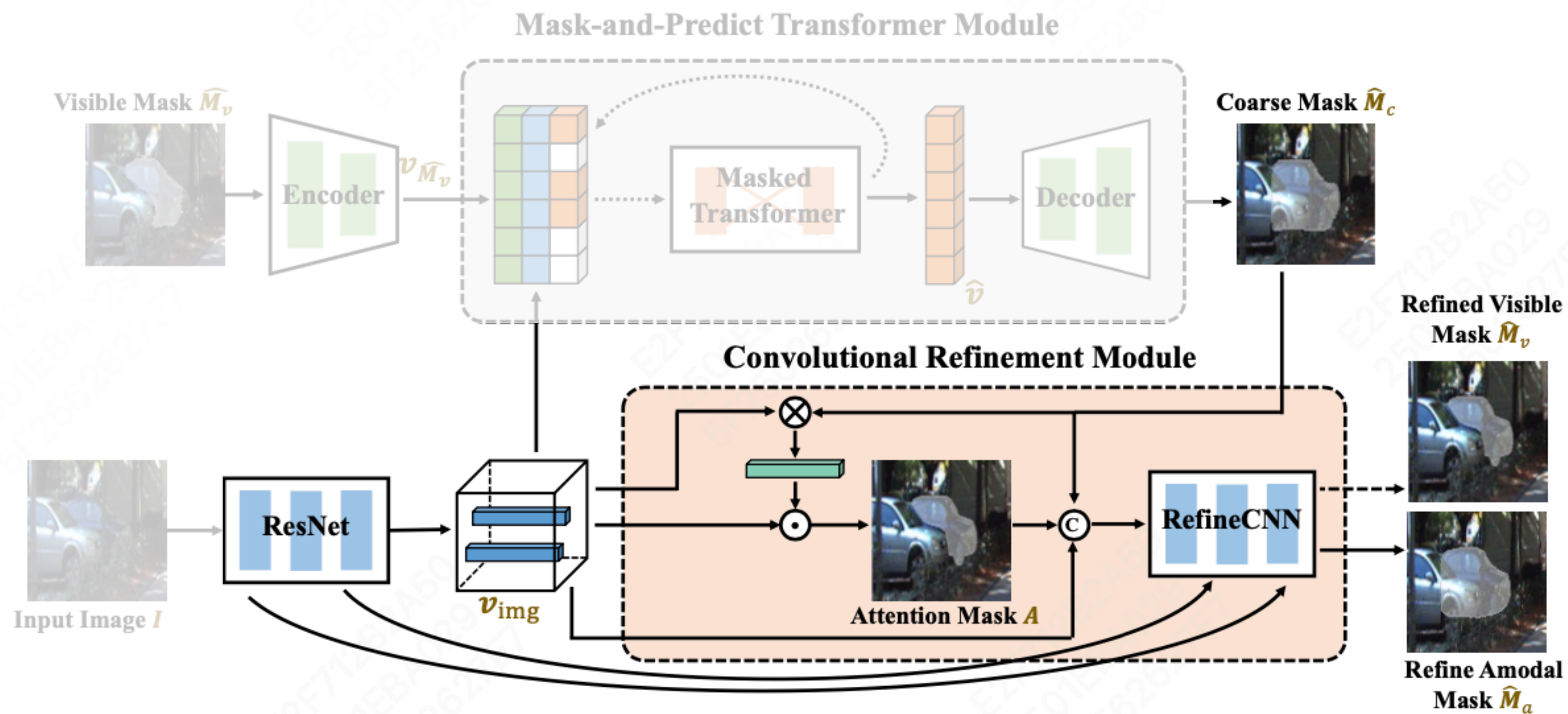
2 转换





3 精修

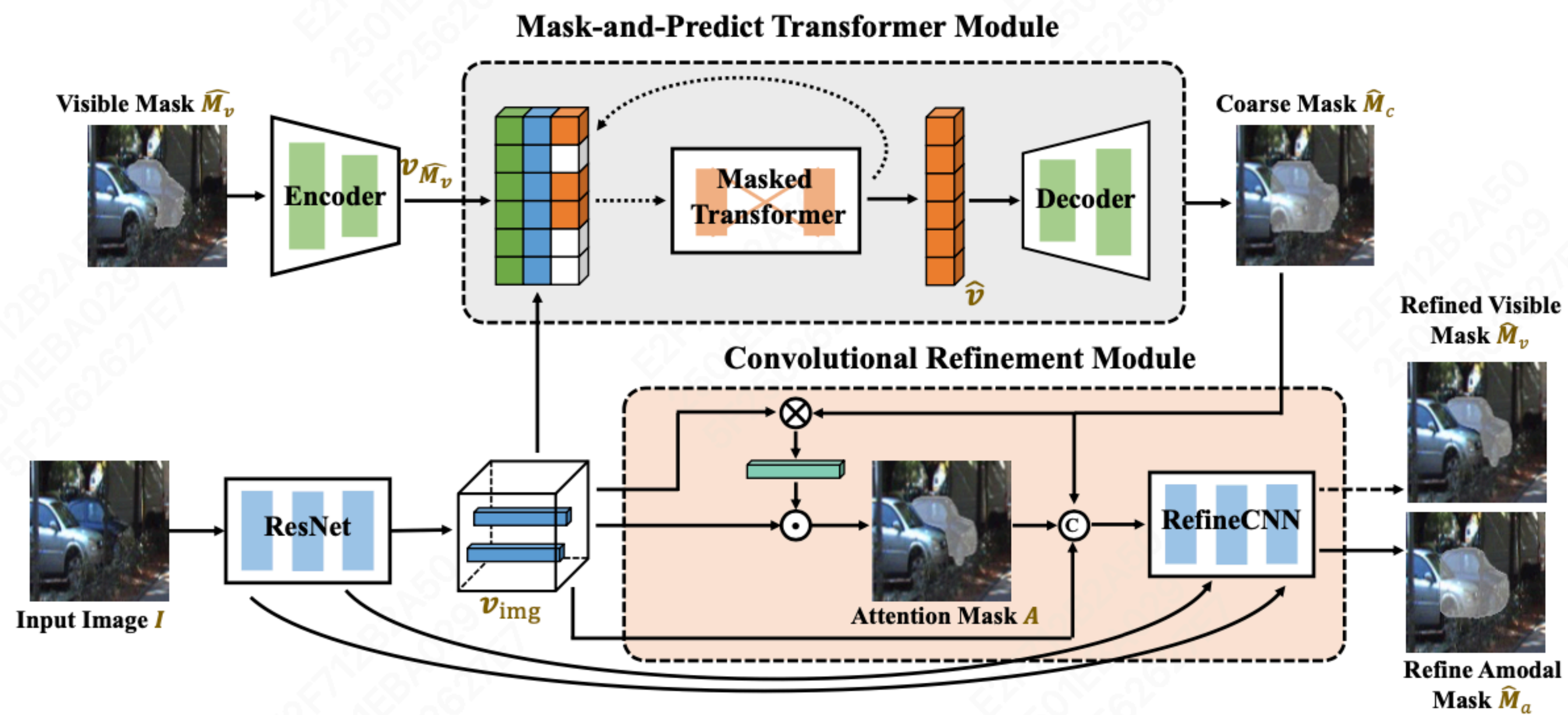
3 精修



$$A := \text{softmax}\left(\frac{\hat{M}_{cd} v_{img}^\top}{\sqrt{d}}\right) \odot v_{img}$$

$$\hat{M}_a, \hat{M}_v = \text{Conv}(v_{img}, A, \hat{M}_{cd})$$

$$\mathcal{L}_r := \text{BCE}(\hat{M}_a, M_a) + \text{BCE}(\hat{M}_v, M_v)$$





数据集

数据集



MOViD-A

合成数据集基于 MOVi

每个场景 10~20 个物体 + 截取 24 帧

训练集 630 组

测试集 208 组



实验

对比



METHODS	KINS						COCOA					
	AP	AP_{50}	AP_{75}	AR	$mIoU_{full}$	$mIoU_{occ}$	AP	AP_{50}	AP_{75}	AR	$mIoU_{full}$	$mIoU_{occ}$
PCNet [35]	29.1	51.8	29.6	18.3	78.02	38.14	-	-	-	-	76.91	20.34
Mask R-CNN [12]	30.0	54.5	30.1	19.4	-	-	28.0	53.7	25.4	29.8	-	-
ORCNN [9]	30.6	54.2	31.3	19.7	-	-	28.0	53.7	25.4	29.8	-	-
VRSP [32]	32.1	55.4	33.3	20.9	80.70	47.33	35.4	56.0	38.7	37.1	78.98	22.92
AISformer [29] [†]	33.8	57.8	35.3	21.1	81.53	48.54	29.0	45.7	31.8	31.1	72.69	13.75
C2F-Seg (<i>ours</i>)	36.5	58.2	37.0	22.1	82.22	53.60	36.6	57.0	38.5	38.5	80.28	27.71

METHODS	FISHBOWL		MOVId-A	
	$mIoU_{full}$	$mIoU_{occ}$	$mIoU_{full}$	$mIoU_{occ}$
<i>visible masks</i>	68.53	-	56.92	-
Convex	77.61	46.38	60.18	16.48
PCNET [35]	87.04	65.02	64.35	27.31
SaVos [34]	88.63	71.55	60.61	22.64
AISformer [29]	-	-	67.72	33.65
C2F-Seg (<i>ours</i>)	91.68	81.21	71.67	36.13

对比



消融



METHODS	KINS		COCOA	
	$mIoU_{full}$	$mIoU_{occ}$	$mIoU_{full}$	$mIoU_{occ}$
w/o refine	81.81	52.57	79.52	24.25
single branch	81.95	52.78	80.03	26.12
full model	82.22	53.60	80.28	27.71

K	COCOA		MOViD-A	
	$mIoU_{full}$	$mIoU_{occ}$	$mIoU_{full}$	$mIoU_{occ}$
1	80.16	27.70	71.91	36.57
2	80.27	27.68	71.67	36.30
3	80.28	27.71	71.67	36.13
5	80.28	27.60	71.58	35.88
8	80.31	27.57	71.46	35.53
10	80.24	27.28	71.42	35.60
12	80.27	27.44	71.41	35.44

消融



GT VM	KINS		COCOA	
	$mIoU_{full}$	$mIoU_{occ}$	$mIoU_{full}$	$mIoU_{occ}$
×	82.22	53.60	80.28	27.71
✓	87.89	57.60	87.13	36.55



写作

摘要



一句话介绍任务 一句话概述方法

Amodal object segmentation is a challenging task that involves segmenting both visible and occluded parts of an object. In this paper, we propose a novel approach, called Coarse-to-Fine Segmentation (C2F-Seg), that addresses this problem by **progressively modeling** the amodal segmentation. C2F-Seg initially reduces the learning space from the pixel-level image space to the vector-quantized latent space. This enables us to better handle long-range dependencies and learn a coarse-grained amodal segment from visual features and visible segments. However, this latent space lacks detailed information about the object, which makes it difficult to provide a precise segmentation directly.

摘要



Amodal object segmentation is a challenging task that involves segmenting both visible and occluded parts of an object. In this paper, we propose a novel approach, called Coarse-to-Fine Segmentation (C2F-Seg), that addresses this problem by progressively modeling the amodal segmentation. C2F-Seg initially reduces the learning space from the pixel-level image space to the **vector-quantized** latent space. This enables us to better handle long-range dependencies and learn a coarse-grained amodal segment from visual features and visible segments.

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简介核心方法

摘要



简介核心方法

数据集

To address this issue, we propose a convolution refine module to inject fine-grained information and provide a more precise amodal object segmentation based on visual features and coarse-predicted segmentation. To help the studies of amodal object segmentation, we create a synthetic amodal dataset, named as MOViD-Amodal (MOViD-A), which can be used for both image and video amodal object segmentation. We extensively evaluate our model on two benchmark datasets: KINS and COCO-A. Our empirical results demonstrate the superiority of C2F-Seg. Moreover, we exhibit the potential of our approach for video amodal object segmentation tasks on FISHBOWL and our proposed MOViD-A. Project page at: <https://jianxgao.github.io/C2F-Seg>.

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实验结果领先

序言



Amodal instance segmentation [24] aims to extract complete shapes of objects in an image, including both visible and occluded parts. This task **plays a vital role** in various real-world applications such as autonomous driving [25, 10], robotics [4], and augmented reality [23, 20]. For instance, in autonomous driving, partial understanding of the scene may result in unsafe driving decisions.

介绍任务 及其重要性

序言



当前的主流方法

Typically, existing approaches [21, 29, 32, 13, 9] build new modules on the detection framework, by additionally introducing an amodal branch that predicts the complete mask perception of the target object. The central idea lies in imbibing a holistic understanding of shape (i.e., shape prior) through multi-task learning by harnessing the supervised signals of the visible and full regions. While these approaches have yielded promising outcomes in recent years, the task of amodal segmentation remains fraught with challenges. One of the main challenges of amodal segmentation is that it is an ill-posed problem, meaning that there are many non-unique and reasonable possibilities for perceiving occluded areas, particularly for elastic bodies like people and animals. On the other hand, there are intricate categories and shapes of objects in real-world scenarios, which would pose significant challenges to prior learning of shapes.

序言



Typically, existing approaches [21, 29, 32, 13, 9] build new modules on the detection framework, by additionally introducing an amodal branch that predicts the complete mask perception of the target object. The central idea lies in imbibing a holistic understanding of shape (i.e., shape prior) through multi-task learning by harnessing the supervised signals of the visible and full regions. **While** these approaches have yielded promising outcomes in recent years, the task of amodal segmentation **remains fraught with challenges. One** of the main challenges of amodal segmentation is that it is an ill-posed problem, meaning that there are many non-unique and reasonable possibilities for perceiving occluded areas, particularly for elastic bodies like people and animals. **On the other hand**, there are intricate categories and shapes of objects in real-world scenarios, which would pose significant challenges to prior learning of shapes.

分析面临挑战

序言



引入本文方法

In this paper, we advocate that shape priors are essential for amodal segmentation, since the shape of an object is usually determined by its function, physiology, and characteristics. For example, a carrot has a long shape, while an apple has a round shape. Thus, the potential distribution of this object can be learned via neural networks. **Nevertheless**, we **argue** that while shape prior can only provide a basic outline and may not capture individual differences or highly local information. Meanwhile, it is possible for the shape prior to being inconsistent with the observed visible area due to factors like pose and viewpoint. To this end, we in this paper propose to generate amodal segments progressively via a coarse-to-fine manner. Specifically, we divide the segmentation of amodal object into two phases: a coarse segmentation phase where we use the shape prior to generate a plausible amodal mask, and a refinement phase is adopted to refine the coarse amodal mask to get the precise segmentation.

序言



In this paper, we advocate that shape priors are essential for amodal segmentation, since the shape of an object is usually determined by its function, physiology, and characteristics. For example, a carrot has a long shape, while an apple has a round shape. Thus, the potential distribution of this object can be learned via neural networks. Nevertheless, we argue that while shape prior can only provide a basic outline and may not capture individual differences or highly local information. Meanwhile, it is possible for the shape prior to being inconsistent with the observed visible area due to factors like pose and viewpoint. **To this end**, we in this paper propose to generate amodal segments progressively via a coarse-to-fine manner. Specifically, we divide the segmentation of amodal object into two phases: a coarse segmentation phase where we use the shape prior to generate a plausible amodal mask, and a refinement phase is adopted to refine the coarse amodal mask to get the precise segmentation.

本文方法概述

序言



In the coarse segmentation phase, as we only need to provide a coarse mask, we perform the segmentation in a low-dimension vector-quantized latent space **to reduce** the learning difficulty and **accelerate** the inference process. The segmentation in such latent space is resorted to the popular mask prediction task adopted in BERT [15] and MaskGIT [3]. Specifically, we adopt a transformer model which takes as inputs the ResNet visual feature, the vector-quantized visible segments, and the ground-truth amodal segments masked in a high ratio. Then the transformer is trained to reconstruct the masked tokens of the amodal segments. This mask-and-predict procedure [3] leads to natural sequential decoding in the inference time. Starting with an all-mask token sequence of amodal segments, our transformer gradually completes the amodal segments. Each step increasingly preserves the most confident prediction.

粗分割阶段介绍

序言



In the second refinement phase, our model learns to inject details to the coarse-prediction and provide a more precise amodal object segmentation. Our convolutional refinement module takes as inputs the coarse-predicted segments and the visual features. **Imitating the human activity for visual stimulus**, we construct a semantic-inspired attention module as an initial stimulus, and then gradually inject the visual features to the segments through convolution layers.

细分割阶段介绍

序言



With this coarse-to-fine architecture **design**, our C2F-Seg complements the latent space that is easier-to-learn, the transformer that has superiority of long-range dependency, and the convolutional model that can supplement details, and results in a better amodal object segmentation. Our framework is flexible to generalize to video-based amodal object segmentation tasks. Guided by the shape prior and visual features of related frames, our model can generate precise amodal segments, and is even capable of generating amodal segments when the object is totally invisible, as shown in Figure 7.

强调各种优点

序言



In order to evaluate the performance of our C2F-Seg. We conduct experiments both on image and video amodal segmentation benchmarks. For image amodal segmentation, our model reaches 36.5/36.6 on AP, 82.22/80.27 on full mIoU and 53.60/27.71 on occluded mIoU for KINS and COCOA respectively. For video amodal segmentation, our model reaches 91.68/71.30 on full mIoU and 81.21/36.04 on occluded mIoU for FISHBOWL and MOVId-A respectively. C2F-Seg **outperforms** all the baselines and **achieves** state-of-the-art performance.

实验结果出色

序言



Our contributions can be summarized as:

- We propose a novel coarse-to-fine **framework**, which consists of a mask-and-predict transformer module for coarse masks and a convolutional refinement module for refined masks. It imitates human activity and progressively generates amodal segmentation, mitigating the effect of detrimental and ill-posed shape priors.
- We build a synthetic **dataset** MOViD-A for amodal segmentation, which contains 838 videos and 12,299 objects. We hope it will advance research in this field. We release the dataset on our project page.
- Extensive experiments are conducted on two image-based benchmarks, showing the **superiority** of our methods over other competitors. Moreover, our framework can be easily extended to video-based amodal segmentation, achieving **state-of-the-art** performance on two benchmarks.

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